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Do Wall Street Landlords Undermine Renters' Welfare?

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1 Big picture

- **Question.** Do institutional landlords in single-family rentals harm or help renters when they expand through mergers?
- **Core setting.** Three large institutional landlord mergers from 2015 to 2017.
- **Treatment.** Overlapped neighborhoods are local areas where both merging firms owned properties before the merger.
- **Main result.** Rents rise after mergers in overlapped neighborhoods, while crime falls.
- **Interpretation.** Market power raises rents, while scale economies and internalization can improve neighborhood safety.

2 Data and measurement

The paper combines institutional property portfolios, Zillow rent data, property-level rental listings, census-tract crime data, security job postings, nightlight data, resident turnover, evictions, vacancy rates, and home values. Treatment intensity is measured by whether the acquirer and target both own properties locally and by Δ Properties, the smaller of the acquirer and target property counts.

3 Models and empirical specifications

3.1 Demand and competition model

Rental demand depends on own rent, rival rent, and neighborhood safety. Mergers raise market power and may raise rent; mergers also raise scale and may increase safety investment.

3.2 Rent DiD

$$\ln(ZRI)_{i,m,t} = \alpha_i + \gamma_{county \times t} + \beta Post_{m,t} \times Treated_{i,m} + \varepsilon_{i,m,t}.$$

3.3 Crime DiD

$$\ln(Crime)_{k,m,t} = \alpha_k + \gamma_{county \times t} + \beta Post_{m,t} \times Treated_{k,m} + \varepsilon_{k,m,t}.$$

3.4 Property-level rent model

$$\ln(Rent)_{p,t} = \alpha_p + \gamma_{neighborhood \times t} + \beta Post_t \times MergedLandlordProperty_p + \varepsilon_{p,t}.$$

4 Identification and empirical design

The design compares overlapped and nonoverlapped neighborhoods around the same mergers. Neighborhood or census-tract fixed effects absorb time-invariant local differences. County-by-time fixed effects absorb county-level shocks. Event-study figures support parallel trends and show treatment effects emerging after merger completion rather than before completion.

5 Tables and figures

5.1 Tables 1–3: Mergers, summary statistics, and treated-control example

Read: Table 1 lists the three mergers. Table 2 summarizes neighborhood, rent, and crime data. Table 3 gives an example of adjacent treated and control neighborhoods.

Detailed interpretation. These tables define the empirical treatment. Overlap is a local consequence of national mergers, not a new local acquisition.

Economic message. The merger design creates local variation in scale and market share.

Table 1
Summary of mergers in the analysis

Name	Status	Acquirer		Name	Status	Target		Announcement	Completion
		Properties	States			Properties	States		
American Homes 4 Rent	Public	29,637 (38,377)	AZ, CA, CO, FL, GA, IL, IN, NV, NC, OH, SC, TX, WA	American Residential Properties	Public	5,247 (8,938)	AZ, CA, CO, FL, GA, IL, IN, NV, NC, OH, SC, TX, WA	Dec. 3, 2015	Feb. 29, 2016
Sturwood Waypoint Residential Trust	Public	10,608 (30,000)	AZ, CA, CO, FL, GA, IL, IN, NV, NC, OH, SC, TX, WA	Colony American Homes	Private	15,537	AZ, CA, CO, FL, GA, IL, IN, NV, NC, TX	Sep. 21, 2015	Jan. 9, 2016
Innovation Homes	Public	42,601 (50,000)	AZ, CA, CO, FL, GA, IL, IN, NV, NC, SC, WA	Sturwood Waypoint Homes	Public	28,491 (32,000)	AZ, CA, CO, FL, GA, IL, IN, NV, NC, OH, SC, TX, WA	Aug. 10, 2017	Nov. 16, 2017

This table summarizes the three horizontal mergers of institutional SFR investors. The numbers of properties are based on the records we manually find from Zillow database. The numbers of properties in parentheses are based on news articles (see Lane 2015a, 2015b, 2017).

Table 2
Summary statistics

A. Neighborhood-level statistics						
Statistic:	N	Mean	P1	P50	P99	SD
Area (square mile)	1,439	3.469	0.170	1.329	40.011	7.238
Properties (per merger)	2,556	10,000	1,000	3,000	103,000	30,467
Δ Properties (per merger)	2,556	1,444	0.000	0.000	23,000	6,675
HHI	2,532	0.581	0.173	0.510	1.000	0.279
Number of rivals	2,556	2.312	0.000	2.000	8.000	2.191
B. Census tract-level statistics						
Area (square mile)	5,603	5.237	0.188	1.237	54.194	30.616
Properties (per merger)	9,708	6.044	1.000	3.000	47.000	10.247
Δ Properties (per merger)	9,708	0.710	0.000	0.000	9.000	2.072
HHI	9,503	0.644	0.200	0.557	1.000	0.274
Number of rivals	9,708	1.643	0.000	1.000	6.000	1.596
C. Market share						
Census tract level:						
Large treatment (premerger)	278	0.027	0.004	0.020	0.145	0.027
Large treatment (postmerger)	278	0.059	0.010	0.047	0.246	0.044
Small treatment (premerger)	1,039	0.011	0.001	0.006	0.096	0.018
Small treatment (postmerger)	1,039	0.027	0.003	0.018	0.132	0.030
Census block level:						
Large treatment (premerger)	17,050	0.132	0.000	0.058	1.000	0.222
Large treatment (postmerger)	17,050	0.234	0.015	0.125	1.000	0.273
Small treatment (premerger)	29,022	0.160	0.000	0.059	1.000	0.270
Small treatment (postmerger)	29,022	0.261	0.012	0.130	1.000	0.306

(Continued)

Panels A and B also report summary statistics on the number of properties owned by merging firms and the number of properties gained from mergers. The average (median) numbers of properties are 10 (3) per neighborhood and 6 (3) per census tract. In the sample, 26.4% (13.6%) of neighborhoods (census tracts) have properties owned by both acquirers and targets prior to the mergers.

Panel C reports the acquirers' market shares before and after the mergers. We first calculate the market share at the census tract level based on the number of properties owned by the acquirer divided by the total number of rental housing units for the census tract reported by the Census Bureau. We also divide overlapped neighborhoods into a "large treatment" group and a "small treatment" group. This division refers to census tracts where the acquirer gained more than five properties and up to five properties after the mergers, respectively. The statistics in the first row show that, at the census tract level, the acquirer's average market share is 1.1% (2.7%) in the small (large) treatment areas prior to the mergers, and it increased to 2.7% (5.9%) after the mergers. These statistics likely understate the true market share of institutional landlords for several reasons. First, the number of rental units reported by the Census Bureau includes non-SFR properties, such as multifamily properties and apartment complexes, which account for 65% of the total rental housing stock (Eisfeldt and Demers 2015) and might not be competing with institutional landlords for the same pool of renters. Second, the total number of rental housing

(a) Summary of mergers

(b) Summary statistics, part 1

Table 2
(Continued)

<i>D. Rent (ZRI) at Zillow neighborhood level</i>						
Period:	Premerger			Postmerger		
Statistic:	N	Mean	SD	N	Mean	SD
American Home 4 Rent	6,444	1,396	443	6,444	1,449	467
American Residential Properties	2,124	1,245	337	2,124	1,286	349
Starwood Waypoint	4,536	1,546	513	4,536	1,668	568
Colony American Homes	6,552	1,676	601	6,552	1,805	661
Colony Starwood Homes	9,588	1,787	656	9,588	1,859	670
Invitation Homes	9,552	1,740	757	9,552	1,800	737
All	30,696	1,639	656	30,696	1,718	675
$I(\Delta Properties > 0)$	8,100	1,617	556	8,100	1,697	577
$I(\Delta Properties > 5)$	1,716	1,661	540	1,716	1,747	562
<i>E. Number of crimes at census tract level</i>						
American Home 4 Rent	26,568	12.40	17.50	26,568	12.61	17.87
American Residential Properties	8,868	15.98	22.57	8,868	11.51	21.12
Starwood Waypoint	16,380	11.12	21.00	16,380	13.34	22.88
Colony American Homes	23,844	11.82	18.77	23,844	11.42	18.78
Colony Starwood Homes	35,256	11.98	20.54	35,256	12.06	20.36
Invitation Homes	35,916	11.43	21.06	35,916	11.09	20.50
All	116,688	11.86	19.68	116,688	11.77	19.78
$I(\Delta Properties > 0)$	15,816	13.10	20.53	15,816	12.74	19.78

(a) Summary statistics, part 2

Table 3
Illustration of treated and control neighborhoods around mergers

Neighborhood	Number of acquirer properties	Number of target properties	Properties gained from merger	Group
North Mountain	56	18	18	Large treatment
Paradise Valley	43	17	17	Large treatment
Deer Valley	42	53	42	Large treatment
Alhambra	16	19	16	Large treatment
Maryvale	5	84	5	Small treatment
Laveen	3	64	3	Small treatment
Estrella	1	60	1	Small treatment
Camelback East	9	2	2	Small treatment
Ahwatukee Foothills	1	3	1	Small treatment
South Mountain	0	32	0	Control
Desert View	0	2	0	Control
North Gateway	0	2	0	Control
North Scottsdale	0	1	0	Control

The table shows the number of properties owned by the acquirer and target in 13 adjacent Zillow Neighborhoods of Maricopa County, Arizona, in the merger between Starwood Waypoint and Colony American Home to illustrate the definition of treated and control neighborhoods. A neighborhood is defined as a "large treatment" if the

(b) Treated/control example

5.2 Figure 1, Table 4, and Figure 2: Rent around mergers

Read: Figure 1 shows the event window. Table 4 shows positive rent effects. Figure 2 shows rent increases only after merger completion.

Detailed interpretation. This is the main market-power evidence: rents rise where merging landlords gain local overlap.

Economic message. Consolidation raises neighborhood rent.

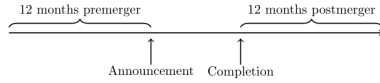


Figure 1
Time line for the difference-in-differences analysis

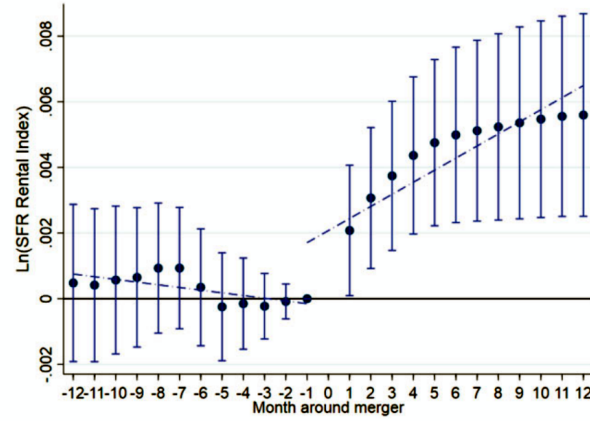
(a) DiD timeline

Table 4
Change in neighborhood rent around mergers

Dependent variable:	ln(ZRI)		
Treatment variable:	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post × Treated	0.0043*** (3.18)	0.0051** (2.55)	0.0002*** (2.95)
Post	-0.0030*** (-4.66)	-0.0013*** (-3.07)	-0.0011*** (-2.87)
Treated	-0.0069*** (-5.31)	-0.0039** (-2.37)	-0.0001 (-1.36)
County × Year-month FEs	Yes	Yes	Yes
Neighborhood FEs	Yes	Yes	Yes
Adjusted R^2	.997	.997	.997
Observations	61,239	61,239	61,239

This table presents estimates of DiD regressions from a sample of neighborhood-month observations from 12 months before the announcement to 12 months after completion of the three horizontal mergers of institutional investors. The sample includes neighborhoods covered by either of the merging firms. We exclude observations between the announcement and completion dates. The dependent variable is the natural logarithm of the Zillow Neighborhood Rental Index (ZRI) for single-family residences. In column 1, *Treated* is a binary variable that equals one if the merging firms gain at least one property after the merger. In column 2, *Treated* is a binary variable that equals one if the merging firms gain more than five properties after the merger. In column 3, *Treated* is the number of properties the merging firms gain after the merger. *Post* is a binary variable that equals one after completion of the mergers. We include county × year-month fixed effects and neighborhood fixed effects in the regressions. We report *t*-statistics using standard errors clustered by neighborhood in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Rent DiD



Rent event study

5.3 Table 5 and Figure 3: Crime around mergers

Read: Crime falls in treated census tracts, and the event study shows declines after completion.

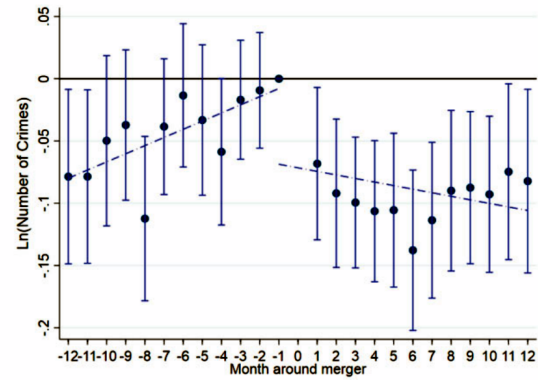
Detailed interpretation. This supports the scale-economy or neighborhood-quality channel.

Economic message. Merged landlords raise rents but may improve safety.

Table 5
Change in neighborhood crime rate around mergers

A. All crimes included						
Dependent variable:	ln(Crime)					
Treatment variable:	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)			
Post $\times$ Treat	-0.0108 (-0.94)	-0.0523** (-2.57)	-0.0048* (-2.91)			
Post	-0.0052 (-1.21)	-0.0025 (-0.78)	-0.0004 (-0.11)			
Treated	-0.0092 (-0.63)	0.0297 (1.29)	0.0026 (1.39)			
County $\times$ Year-month FEs	Yes	Yes	Yes			
Census tract FEs	Yes	Yes	Yes			
Adjusted R^2	.886	.886	.886			
Observations	232,397	232,397	232,397			
B. Crimes by type						
Dependent variable:	ln(Crime)					
Type of crime:	Assault (1)	Burglary (2)	Robbery (3)	Theft (4)	Drug (5)	Vandalism (6)
Post $\times$ Treat	-0.0198 (-1.60)	-0.0303* (-1.81)	-0.0143 (-1.63)	-0.0464*** (-2.59)	-0.0118 (-1.04)	-0.0343*** (-2.75)
Post	0.0003 (0.18)	0.0006 (0.27)	0.0014 (1.03)	-0.0006 (-0.24)	-0.0006 (-0.42)	-0.0012 (-0.73)
Treated	0.0143 (1.21)	0.0079 (0.53)	0.0130** (2.00)	0.0299* (1.77)	0.0042 (0.42)	-0.0125 (-1.03)
County $\times$ Year-month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	.730	.753	.404	.803	.654	.623
Observations	232,397	232,397	232,397	232,397	232,397	232,397

This table presents estimates of DiD regressions from a sample of census-tract-month observations from 12 months before the announcement to 12 months after completion of the three horizontal mergers of institutional investors. The sample includes census tracts owned by either of the merging firms. We exclude observations



(a) Crime DiD

(b) Crime event study

5.4 Tables 6–10: Mechanism tests

Read: Target-property tests, portfolio correlations, security-related amenities, turnover, and eviction evidence explain how rent and crime change.

Detailed interpretation. Amenities and lower turnover support a quality mechanism, while lack of eviction increase reduces concern that crime falls because of aggressive evictions.

Economic message. The quality channel is visible beyond crime itself.

Table 6
Target properties and postmerger changes in rent and crime rate

Dependent variable:	ln(ZRI) (1)	ln(Crime) (2)
Post × Target properties (nonoverlapped)	0.0000 (0.52)	−0.0019* (−1.88)
Post × Target properties (overlapped)	0.0001*** (5.87)	−0.0031*** (−3.85)
Target properties (nonoverlapped)	0.0000 (0.03)	0.0037*** (3.19)
Target properties (overlapped)	0.0000 (0.84)	0.0023** (2.57)
Post	−0.0008** (−2.27)	0.0064 (1.51)
County × Year-month FEs	Yes	Yes
Neighborhood FEs	Yes	
Census tract FEs		Yes
Adjusted R^2	.997	.887
Observations	61,239	232,397

This table shows that the postmerger increase (decrease) in rent (crime rate) is more correlated with the number of target properties in overlapped neighborhoods than in nonoverlapped neighborhoods. Column 1 presents estimates of DiD regressions from a sample of neighborhood-month observations from 12 months before the announcement to 12 months after completion of the three horizontal mergers of institutional investors. The sample includes neighborhoods covered by either of the merging firms. The dependent variable is the natural logarithm

(a) Target properties

Table 7
Historical correlation between acquirer's and target's portfolios

Merger	Distribution of correlations within counties					Overall correlation
	Min (1)	p25 (2)	p50 (3)	p75 (4)	Max (5)	
American Homes 4 Rent – American Residential Properties	.150	.206	.263	.596	.884	.356
Starwood Waypoint Residential Trust – Colony American Homes	.190	.589	.741	.839	.902	.958
Invitation Homes – Starwood Waypoint Homes	.002	.204	.483	.727	.891	.774

This table shows the historical correlation in rent between the acquirer's and the target's portfolios. Columns 1 to 5 show the distribution of correlation at the county level. Specifically, within each county, we calculate the average rent growth for the acquirer's and the target's portfolios using the first difference of neighborhood-level $\ln(ZRI)$ weighted by the number of properties in each neighborhood. We then calculate the correlation using the weighted-average rent growth over the 36 months prior to the merger announcements. In column 6, we calculate the historical correlation using the weighted-average rent growth across the entire acquirer's and target's portfolios.

is likely greater than the average treatment effect for a hypothetical person in

(b) Portfolio correlation

Table 8
Change in neighborhood amenities around mergers

Dependent variable:	A Number of job posts for security guards		
	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post × Treated	8.5742** (2.30)	9.5413** (2.23)	0.0319** (2.55)
Post	−5.0386*** (−2.84)	−4.7936*** (−2.74)	−0.4450 (−0.32)
Treated	−4.7184* (−1.70)	−4.3365 (−1.51)	0.0217 (1.40)
County FEs	Yes	Yes	Yes
Year-month FEs	Yes	Yes	Yes
Adjusted R^2	.807	.807	.813
Observations	10,254	10,254	10,254

Dependent variable:	B ln(streetlight radiance)		
	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post × Treated	0.0059** (1.98)	0.0084 (1.09)	0.0010* (1.92)
Post	−0.0016* (−1.79)	−0.0007 (−1.03)	−0.0014* (−1.79)
Treated	−0.0072*** (−2.79)	−0.0129*** (−2.85)	−0.0013*** (−3.03)
County × Year-month FEs	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes
Adjusted R^2	.989	.989	.989
Observations	66,437	66,437	66,437

Panel A presents estimates of DiD regressions from a sample of county-year-month observations from 12 months before the announcement to 12 months after completion of the three horizontal mergers of institutional investors. The sample includes counties covered by either of the merging firms and exclude observations between the announcement and completion dates. The dependent variable is the number of job posts for security guards in the county. We include county fixed effects and year-month fixed effects, and report t -statistics using standard errors clustered by county in parentheses. Panel B presents estimates of DiD regressions from a sample of

(a) Amenities

Table 9
Change in resident turnover around mergers

Dependent variable:	ln(turnover)		
Treatment variable:	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post × Treated	−0.0326*** (−4.75)	−0.0570*** (−2.99)	−0.0075*** (−3.40)
Post	−0.0202*** (−7.49)	−0.0333*** (−14.30)	−0.0191*** (−5.90)
Treated	0.1188*** (15.93)	0.2309*** (12.60)	0.0256*** (9.97)
County × Year-month FEs	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes
Adjusted R^2	.303	.304	.307
Observations	232,397	232,397	232,397

This table presents estimates of DiD regressions from a sample of census-tract-month observations from 12 months before the announcement to 12 months after completion of the three horizontal mergers of institutional investors. The sample includes census tracts covered by either of the merging firms. We exclude observations between the announcement and completion dates. In panel A, the dependent variable is the natural logarithm of one plus the number of renter turnovers for properties owned by the merged landlords. In column 1, *Treated* is a binary variable that equals one if the merging firms gain at least one property after the merger. In column 2, *Treated* is a binary variable that equals one if the merging firms gain more than five properties after the merger. In column 3, *Treated* is the number of properties the merging firms gain after the merger. *Post* is a binary variable that equals one after completion of the mergers. We include county × year-month fixed effects and census tract fixed effects in the regressions. We report t -statistics using standard errors clustered by census tract in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Turnover

Table 10
Change in eviction rate around mergers

Dependent variable:	Eviction rate		
Treatment variable:	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post $\times$ Treated	0.1311 (0.93)	0.1073 (0.26)	0.0199 (0.61)
Treated	-0.0732 (-0.93)	-0.0558 (-0.26)	-0.0108 (-0.61)
County $\times$ Year FEs	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes
Adjusted R^2	.817	.817	.817
Observations	5,032	5,032	5,032

This table presents estimates of DiD regressions from a sample of census-tract-year observations from 1 year before the announcement to 1 year after completion of the first two horizontal mergers of institutional investors. The sample includes census tracts covered by either of the merging firms. We exclude the year of merger announcement. The dependent variable is the eviction rate (in percentage points) for a census tract reported by the Eviction Lab. In column 1, *Treated* is a binary variable that equals one if the merging firms gain at least one property after the merger. In column 2, *Treated* is a binary variable that equals one if the merging firms gain more

Eviction rate

5.5 Tables 11–12: Local competition

Read: Effects are moderated by local competition in rent and crime specifications.

Detailed interpretation. Competition tests support the market power interpretation.

Economic message. Merger effects are stronger when local competition is weaker.

Table 11
Local market competition and change in neighborhood rent around mergers

A			
Dependent variable:	ln(ZRI)		
Treatment variable:	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post $\times$ Treated	0.0037** (2.48)	0.0057** (2.57)	0.0003*** (4.10)
Post	-0.0032*** (-4.83)	-0.0017*** (-3.81)	-0.0017*** (-3.88)
Treated	-0.0065*** (-4.10)	-0.0044** (-2.30)	-0.0001 (-1.02)
HHI	0.0004 (0.05)	-0.0039 (-0.63)	-0.0055 (-0.90)
Post $\times$ HHI	-0.0099*** (-3.71)	-0.0102*** (-4.22)	-0.0102*** (-4.25)
Treated $\times$ HHI	-0.0047 (-0.80)	-0.0163** (-2.35)	-0.0003 (-1.11)
Post $\times$ Treated $\times$ HHI	0.0052 (1.00)	0.0195** (2.36)	0.0010*** (3.25)
County $\times$ Year-month FEs	Yes	Yes	Yes
Neighborhood FEs	Yes	Yes	Yes
Adjusted R^2	.997	.997	.997
Observations	60,681	60,681	60,681
B			
Post $\times$ Treated	0.0023 (1.04)	0.0065 (1.55)	0.0005** (1.98)
Post	-0.0073*** (-7.22)	-0.0064*** (-7.64)	-0.0064*** (-7.58)
Treated	-0.0076*** (-3.17)	-0.0086** (-2.32)	-0.0007*** (-3.01)
Number of rivals	0.0005 (0.57)	0.0016** (2.02)	0.0014* (1.66)
Post $\times$ Number of rivals	0.0022** (6.50)	0.0024*** (7.77)	0.0022*** (7.62)
Treated $\times$ Number of rivals	0.0009 (1.55)	0.0018** (2.50)	0.0001*** (3.24)
Post $\times$ Treated $\times$ Number of rivals	-0.0002 (-0.32)	-0.0016* (-1.90)	-0.0001** (-2.04)
County $\times$ Year-month FEs	Yes	Yes	Yes
Neighborhood FEs	Yes	Yes	Yes
Adjusted R^2	.997	.997	.997
Observations	61,239	61,239	61,239

(Continued)

Table 11
(Continued)

C			
Dependent variable:	ln(ZRI)		
Treatment variable:	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post $\times$ Treated	0.0038*** (2.72)	0.0042** (2.01)	0.0002* (1.72)
Post	-0.0024*** (-3.39)	-0.0007 (-1.39)	-0.0009* (-1.91)
Treated	-0.0066*** (-4.72)	-0.0034* (-1.91)	-0.0002*** (-3.27)
Market share	0.0236* (1.67)	0.0181 (1.63)	0.0050 (0.69)
Post $\times$ Market share	-0.0240*** (-3.71)	-0.0197*** (-3.28)	-0.0130** (-2.31)
Treated $\times$ Market share	-0.0179 (-1.38)	-0.0159 (-1.54)	0.0002** (2.08)
Post $\times$ Treated $\times$ Market share	0.0232*** (3.33)	0.0215*** (3.19)	0.0002* (1.69)
County $\times$ Year-month FEs	Yes	Yes	Yes
Neighborhood FEs	Yes	Yes	Yes
Adjusted R^2	.997	.997	.997
Observations	61,239	61,239	61,239

This table presents estimates of DiD regressions from a sample of neighborhood-month observations from 12 months before the announcement to 12 months after completion of the three horizontal mergers of institutional investors. The sample includes neighborhoods covered by either of the merging firms. We exclude observations between the announcement and completion dates. The dependent variable is the natural logarithm of the Zillow Neighborhood Rental Index (ZRI) for single-family residences. In column 1, *Treated* is a binary variable that equals one if the merging firms gain at least one property after the merger. In column 2, *Treated* is a binary variable that equals one if the merging firms gain more than five properties after the merger. In column 3, *Treated* is the number of properties the merging firms gain after the merger. *Post* is a binary variable that equals one after completion of the mergers. *HHI* is the (demeaned) Herfindahl-Hirschman index measuring the concentration of institutional SFR investors in the neighborhood. *Number of rivals* is the number of rival institutional SFR investors in the neighborhood. *Market share* is the postmerger market share, defined as the ratio of the number of properties owned by the merged landlord to the imputed number of rental units for the neighborhood. We include county $\times$ year-month fixed effects and neighborhood fixed effects in the regressions. We report *t*-statistics using standard errors clustered by neighborhood in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Rent competition, part 1

(b) Rent competition, part 2

Table 12
Local market competition and change in neighborhood crime rate around mergers

Dependent variable:	A		
	ln(Crime)		
Treatment variable:	$I(\Delta Properties > 0)$	$I(\Delta Properties > 5)$	$\Delta Properties$
	(1)	(2)	(3)
Post $\times$ Treated	-0.0058 (-0.54)	-0.0377** (-2.09)	-0.0040** (-2.49)
Post	0.0047 (0.90)	0.0085* (1.88)	0.0091* (1.88)
Treated	-0.0233 (-1.35)	0.0047 (0.15)	-0.0005 (-0.19)
HHI	0.1786*** (2.74)	0.1627*** (2.66)	0.1795*** (2.85)
Post $\times$ HHI	0.0406 (1.47)	0.0383 (1.47)	0.0366 (1.35)
Treated $\times$ HHI	-0.0250 (-0.38)	-0.1404 (-1.07)	-0.0136 (-1.32)
Post $\times$ Treated $\times$ HHI	0.0146 (0.33)	0.0942 (1.07)	0.0041 (0.56)
County $\times$ Year-month FEs	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes
Adjusted R^2	.886	.886	.886
Observations	227,538	227,538	227,538
Dependent variable:	B		
	ln(Crime)		
Treatment variable:	$I(\Delta Properties > 0)$	$I(\Delta Properties > 5)$	$\Delta Properties$
	(1)	(2)	(3)
Post $\times$ Treated	0.0161 (0.88)	0.0194 (0.49)	0.0019 (0.51)
Post	0.0151 (1.52)	0.0178* (1.90)	0.0156 (1.61)
Treated	-0.0286 (-0.95)	-0.0565 (-0.87)	-0.0074 (-1.18)
Number of rivals	0.0134 (1.17)	0.0152 (1.39)	0.0113 (1.01)
Post $\times$ Number of rivals	-0.0125** (-2.30)	-0.0120** (-2.41)	-0.0114** (-2.22)
Treated $\times$ Number of rivals	0.0062 (0.68)	0.0237 (1.37)	0.0026* (1.71)
Post $\times$ Treated $\times$ Number of rivals	-0.0062 (-0.78)	-0.0164 (-1.16)	-0.0014 (-1.29)
County $\times$ Year-month FEs	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes
Adjusted R^2	.887	.887	.887
Observations	232,397	232,397	232,397

(Continued)

Table 12
(Continued)

Dependent variable:	C		
	ln(Crime)		
Treatment variable:	$I(\Delta Properties > 0)$	$I(\Delta Properties > 5)$	$\Delta Properties$
	(1)	(2)	(3)
Post $\times$ Treated	0.0053 (0.35)	0.0039 (0.11)	-0.0000 (-0.00)
Post	-0.0055 (-1.04)	-0.0043 (-0.96)	-0.0037 (-0.77)
Treated	-0.0173 (-0.93)	0.0016 (0.04)	0.0017 (0.50)
Market share	-0.5537* (-1.66)	-0.5372** (-2.36)	-0.5694** (-2.48)
Post $\times$ Market share	-0.3614 (-1.20)	-0.2763 (-1.18)	-0.2626 (-1.14)
Treated $\times$ Market share	0.5137 (1.22)	0.7259 (1.47)	0.0318 (1.10)
Post $\times$ Treated $\times$ Market share	-0.2485 (-0.60)	-0.7708* (-1.65)	-0.0415** (-2.22)
County $\times$ Year-month FEs	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes
Adjusted R^2	.887	.887	.887
Observations	232,349	232,349	232,349

This table presents estimates of DiD regressions from a sample of census-tract-month observations from 12 months before the announcement to 12 months after completion of the three horizontal mergers of institutional investors. The sample includes census tracts covered by either of the merging firms. We exclude observations between the announcement and completion dates. The dependent variable is the natural logarithm of one plus the number of criminal incidents. In column 1, *Treated* is a binary variable that equals one if the merging firms gain at least one property after the merger. In column 2, *Treated* is a binary variable that equals one if the merging firms gain more than five properties after the merger. In column 3, *Treated* is the number of properties the merging firms gain after the merger. *Post* is a binary variable that equals one after completion of the mergers. *HHI* is the (demeaned) Herfindahl-Hirschman index measuring the concentration of institutional SFR investors in the census tract. *Number of rivals* is the number of rival institutional SFR investors in the census tract. *Market share* is the postmerger market share, defined as the ratio of the number of properties owned by the merged landlord to the number of rental units reported by the Census Bureau. We include county $\times$ year-month fixed effects and census tract fixed effects in the regressions. We report *t*-statistics using standard errors clustered by census tract in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Crime competition, part 1

(b) Crime competition, part 2

5.6 Tables 13–15: Property-level rent, vacancy, and home values

Read: Merged landlords raise property-level rents; vacancy rises; home values do not rise strongly.

Detailed interpretation. The evidence supports real market power rather than pure quality improvements.

Economic message. Rent increases are not fully explained by better neighborhood quality.

Table 13
Change in property-level rent around mergers

A			
Dependent variable:	ln(Rent)		
Treatment variable:	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post	0.0058* (1.77)	0.0049** (2.04)	0.0066*** (2.83)
Treated	-0.0080* (-1.72)	-0.0117*** (-3.59)	-0.0004* (-1.87)
Post $\times$ Treated	-0.0028 (-0.69)	-0.0025 (-0.59)	-0.0005** (-2.12)
Merged landlord	0.0270*** (9.81)	0.0249*** (10.59)	0.0254*** (10.94)
Post $\times$ Merged landlord	-0.0048 (-1.48)	-0.0019 (-0.69)	-0.0018 (-0.68)
Treated $\times$ Merged landlord	-0.0029 (-0.77)	0.0021 (0.52)	0.0000 (0.05)
Post $\times$ Treated $\times$ Merged landlord	0.0119*** (2.59)	0.0093* (1.95)	0.0005** (2.25)
Census tract $\times$ Year-month FEs	Yes	Yes	Yes
Adjusted R^2	.679	.679	.679
Observations	152,427	152,427	152,427
B			
Post	0.0056* (1.72)	0.0051** (2.10)	0.0070*** (2.92)
Treated	-0.0082* (-1.77)	-0.0117*** (-3.60)	-0.0004* (-1.85)
Post $\times$ Treated	-0.0025 (-0.62)	-0.0026 (-0.63)	-0.0005** (-2.23)
Acquirer	0.0290*** (8.94)	0.0252*** (9.23)	0.0256*** (9.64)
Post $\times$ Acquirer	-0.0091** (-2.13)	-0.0076** (-2.19)	-0.0077** (-2.32)
Treated $\times$ Acquirer	-0.0071 (-1.58)	0.0003 (0.07)	-0.0001 (-0.24)
Post $\times$ Treated $\times$ Acquirer	0.0145** (2.49)	0.0153*** (2.60)	0.0009*** (3.15)
Target	0.0230*** (5.07)	0.0243*** (6.47)	0.0252*** (7.07)
Post $\times$ Target	0.0024 (0.49)	0.0073* (1.84)	0.0073* (1.96)
Treated $\times$ Target	0.0048 (0.85)	0.0047 (0.80)	0.0001 (0.46)
Post $\times$ Treated $\times$ Target	0.0074 (1.75)	-0.0002 (-0.03)	0.0000 (0.06)

(a) Property-level rent

Table 14
Change in the vacancy rate around mergers

Dependent variable:	Vacancy rate		
Treatment variable:	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post $\times$ Treated	0.0455** (2.55)	0.0402* (1.82)	0.0026 (1.38)
Post	0.0174*** (2.89)	0.0076* (1.96)	0.0132*** (2.99)
Treated	0.0765*** (5.16)	0.0652*** (3.85)	0.0074*** (5.66)
County $\times$ Year-Quarter FEs	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes
Adjusted R^2	.947	.947	.947
Observations	154,905	154,905	154,905

This table presents estimates of DiD regressions from a sample of census-tract-quarter observations from four quarters before the announcement to four quarters after completion of the three horizontal mergers of institutional investors. The sample includes census tracts covered by either of the merging firms. We exclude observations between the announcement and completion dates. The dependent variable is the vacancy rate for a census tract (in percentage) reported by the U.S. Postal Service. In column 1, *Treated* is a binary variable that equals one if the merging firms gain at least one property after the merger. In column 2, *Treated* is a binary variable that equals one if the merging firms gain more than five properties after the merger. In column 3, *Treated* is the number of properties the merging firms gain after the merger. *Post* is a binary variable that equals one after completion of the mergers. We include county $\times$ year-quarter fixed effects and census tract fixed effects in the regressions. We report *t*-statistics using standard errors clustered by census tract in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Vacancy

Table 15
Change in neighborhood home value around mergers

A. Change in home value 1-year postmerger			
Dependent variable:	ln(ZHVI)		
Treatment variable:	$I(\Delta Properties > 0)$ (1)	$I(\Delta Properties > 5)$ (2)	$\Delta Properties$ (3)
Post $\times$ Treated	0.0026 (1.15)	-0.0008 (-0.23)	0.0001 (1.13)
Post	-0.0041*** (-3.46)	-0.0011 (-1.58)	-0.0008 (-1.30)
Treated	-0.0138*** (-4.99)	-0.0062** (-2.32)	-0.0001 (-1.31)
County $\times$ Year-month FEs	Yes	Yes	Yes
Neighborhood FEs	Yes	Yes	Yes
Adjusted R^2	.995	.995	.995
Observations	68,920	68,920	68,920
B. Change in home value two years postmerger			
Post $\times$ Treated	0.0062** (2.31)	0.0009 (0.21)	0.0003** (2.27)
Post	-0.0045*** (-3.38)	-0.0011 (-1.43)	-0.0009 (-1.33)
Treated	-0.0135*** (-5.01)	-0.0067*** (-3.00)	-0.0002** (-2.44)
County $\times$ Year-month FEs	Yes	Yes	Yes
Neighborhood FEs	Yes	Yes	Yes
Adjusted R^2	.993	.993	.993
Observations	109,038	109,038	109,038

This table presents estimates of DiD regressions in which the dependent variable is the natural logarithm of Zillow Neighborhood Home Value Index (ZHVI) for single-family residences. Panel A (B) uses a sample of neighborhood-month observations from 12 months before the announcement to 12 (24) months after completion of the three horizontal mergers of institutional investors. Both samples include only neighborhoods covered by either of the merging firms and exclude observations between the announcement and completion dates. In column 1, *Treated* is a binary variable that equals one if the merging firms gain at least one property after the merger. In column 2, *Treated* is a binary variable that equals one if the merging firms gain more than five properties after the merger. In column 3, *Treated* is the number of properties the merging firms gain after the merger. *Post* is a binary variable that equals one after completion of the mergers. We include county $\times$ year-month fixed effects and neighborhood fixed effects in the regressions. We report *t*-statistics using standard errors clustered by neighborhood in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Home values

5.7 Table 16, Figure 4, Table 17, and Figure 5: Broad-sample robustness

Read: Broader samples confirm rent increases and crime reductions.

Detailed interpretation. The patterns are not limited to the baseline overlap sample.

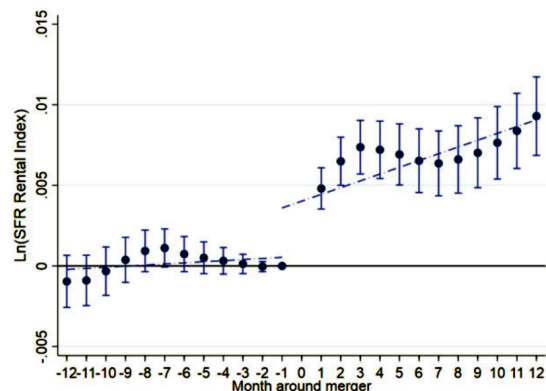
Economic message. The main results are robust across treated-group definitions.

Table 16
Comparing rent between neighborhoods with and without merging firms

Dependent variable:	ln(ZRI)	
	(1)	(2)
Post	-0.0015*** (-4.35)	-0.0017*** (-4.78)
$I(Properties > 0)$	0.0030*** (3.45)	
Post $\times I(Properties > 0)$	0.0069*** (7.04)	
$I(\Delta Properties > 0)$		0.0004 (0.27)
Post $\times I(\Delta Properties > 0)$		0.0095*** (6.44)
$I(Properties > 0 \ \& \ \Delta Properties = 0)$		0.0036*** (4.00)
Post $\times I(Properties > 0 \ \& \ \Delta Properties = 0)$		0.0061*** (5.92)
County $\times$ Year-month FEs	Yes	Yes
Neighborhood FEs	Yes	Yes
Adjusted R^2	.996	.996
Observations	182,932	182,932

This table presents estimates of DiD regressions from a sample of neighborhood-month observations from 12 months before the announcement to 12 months after completion of the three horizontal mergers of institutional investors. The sample includes all neighborhoods (with available data) in the counties covered by either of the merging firms. We exclude observations between the announcement and completion dates. The dependent variable is the natural logarithm of the Zillow Neighborhood Rental Index (ZRI) for single-family residences. $I(Properties > 0)$ is a binary variable that equals one if either of the merging firms has properties in the neighborhood. $I(\Delta Properties > 0)$ is a binary variable that equals one if both of the merging firms have properties in the neighborhood. $I(Properties > 0 \ \& \ \Delta Properties = 0)$ is a binary variable that equals one if only one of the merging firms has properties in the neighborhood. We include county $\times$ year-month fixed effects and neighborhood fixed effects in the regressions. We report t -statistics using standard errors clustered by neighborhood in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Broad rent table



(b) Broad rent event study

Table 17
Comparing crime rate between neighborhoods with and without merging firms

Dependent variable:	A					
	All	Assault	Burglary	Robbery	Theft	Drug
Type of crime:	(1)	(2)	(3)	(4)	(5)	(6)
Post	0.0203*** (7.63)	0.0083*** (6.66)	0.0105*** (7.30)	0.0040*** (6.94)	0.0148*** (7.98)	0.0033*** (3.52)
$I(Properties > 0)$	0.0278*** (3.75)	0.0177*** (4.96)	0.0142*** (4.07)	0.0096*** (4.96)	0.0190*** (3.47)	0.0064*** (2.36)
Post $\times I(Properties > 0)$	-0.0171*** (-8.31)	-0.0248*** (-8.27)	-0.0368*** (-8.09)	-0.0155*** (-6.92)	-0.0532*** (-7.90)	-0.0103*** (-3.45)
County $\times$ Year-month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	.843	.702	.723	.419	.771	.623
Observations	843,240	843,240	843,240	843,240	843,240	843,240

(Continued)

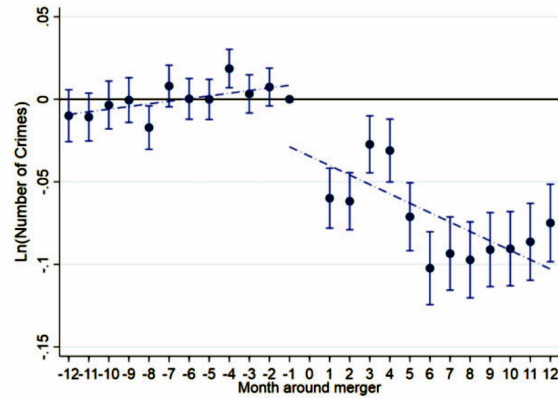
Table 17
(Continued)

Dependent variable:	B					
	All	Assault	Burglary	Robbery	Theft	Drug
Type of crime:	(1)	(2)	(3)	(4)	(5)	(6)
Post	0.0208*** (7.73)	0.0084*** (6.78)	0.0109*** (7.41)	0.0050*** (7.09)	0.0033*** (3.53)	0.0025*** (2.42)
$I(\Delta Properties > 5)$	0.0573*** (2.22)	0.0317*** (2.51)	0.0217*** (1.39)	0.0219*** (2.93)	0.0056*** (2.51)	0.0002 (0.97)
Post $\times I(\Delta Properties > 5)$	-0.1117*** (-4.42)	-0.0418*** (-3.23)	-0.0668*** (-3.69)	-0.0225*** (-2.74)	-0.0891*** (-4.01)	-0.0208*** (-1.84)
$I(Properties > 0 \ \& \ \Delta Properties \leq 5)$	0.0273*** (3.67)	0.0175*** (4.91)	0.0144*** (3.97)	0.0088*** (4.86)	0.0185*** (3.38)	0.0063*** (2.31)
Post $\times I(Properties > 0 \ \& \ \Delta Properties \leq 5)$	-0.0708*** (-8.35)	-0.0254*** (-4.12)	-0.0382*** (-3.89)	-0.0152*** (-4.24)	-0.0252*** (-7.73)	-0.0099*** (-3.35)
County $\times$ Year-month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Census tract FEs	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	.843	.702	.723	.419	.771	.623
Observations	843,240	843,240	843,240	843,240	843,240	843,240

This table presents estimates of DiD regressions from a sample of census tract-month observations from 12 months before the announcement to 12 months after completion of the three horizontal mergers of institutional investors. The sample includes all census tracts (with available data) in the counties covered by either of the merging firms. We exclude observations between the announcement and completion dates. The dependent variable is the natural logarithm of one plus the number of crimes in the census tract. $I(Properties > 0)$ is a binary variable that equals one if either of the merging firms has properties in the census tract. $I(\Delta Properties > 5)$ is a binary variable that equals one if the merging firms gain more than five properties in the census tract after the merger. $I(Properties > 0 \ \& \ \Delta Properties \leq 5)$ is a binary variable that equals one if only one of the merging firms has properties in the census tract or the postmerger gain is equal to less than five properties after the merger. Post is a binary variable that equals one after completion of the merger. We include county $\times$ year-month fixed effects and census tract fixed effects in the regressions. We report t -statistics using standard errors clustered by census tract in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Broad crime, part 1

(b) Broad crime, part 2



Broad crime event study

6 Short summary

- Institutional landlord mergers raise rents in overlapped neighborhoods.
- Crime falls after merger completion in overlapped census tracts.
- Amenities, turnover, and competition tests support the quality and market-power channels.
- Property-level rent and vacancy evidence suggest market power is real.

7 Conclusion

7.1 Core conclusion

Institutional landlord consolidation has two-sided effects: it raises rents but also lowers crime.

7.2 Economic intuition

Market power increases rent, while scale economies and internalization improve neighborhood quality.

7.3 Policy implication

Antitrust and housing policy should evaluate both price and quality margins.

7.4 Limitations

The treatment is not random, ZRI is selective, and the paper does not monetize crime reductions relative to rent increases.

7.5 Takeaway

Wall Street landlords can both extract more surplus and improve local public goods.